

Analysis of River Flooding using Particle Method with PLATEAU

PLATEAUを用いた粒子法による河川氾濫解析

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2026.08.05 AOGS**

I am Yukitake Wakamiya from Nagasaki University.
The title of my presentation is "Analysis of River Flooding using Particle Method with PLATEAU."

Prediction of Damage Risk



Exposed
Population



Behavior of
inundated flows



Building
Damage

etc.

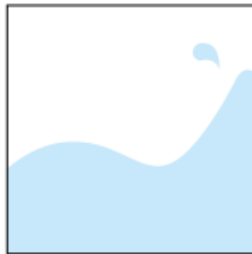
Damage Risk Prediction by River Inundation Analysis
Increasingly Needed in Recent Heavy Rainfall Disasters

First, the background and purpose.

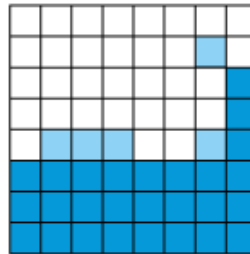
In recent years, heavy rainfall disasters have occurred frequently in Japan. By conducting more detailed river inundation analysis, we need to predict various damage risks.

This includes investigating the exposed population, the behavior of inundated flows, and damage to buildings.

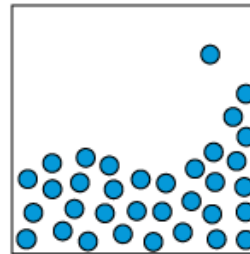
Types of flooding analysis



Actual water surface



Grid method



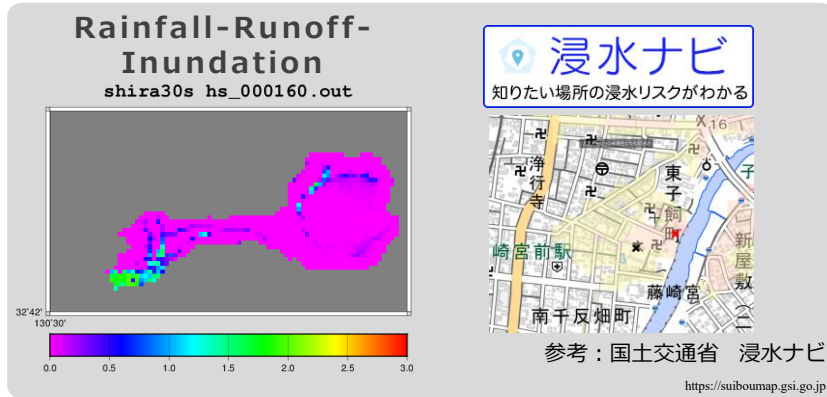
Particle method

FYI : <https://qiita.com/KionIm/items/02b1aadbfa38b11c3b83>

There are two main methods for river flood analysis: the grid method and the particle method.

The grid method divides the area into cells, and solves the flow cell by cell. The particle method treats the water as a set of moving particles, and calculates how they interact with each other.

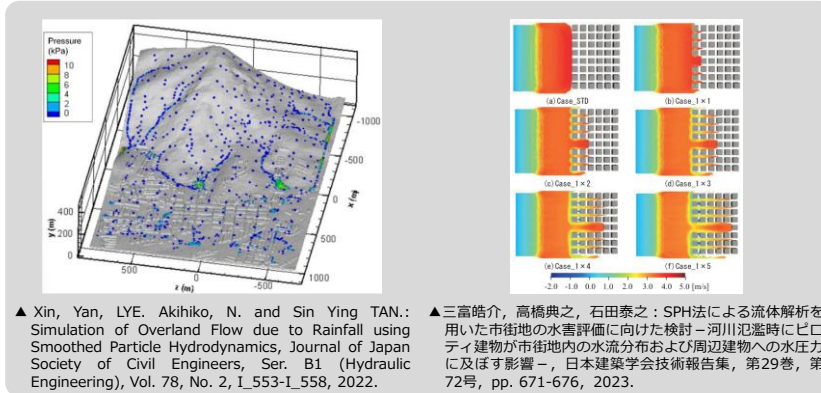
Grid method



A method used in many river inundation analyses, in which land is divided into parcels and analysis is performed on a parcel-by-parcel basis

The grid method is commonly used for river flood analysis in Japan, for example in the RRI model and in flood hazard maps.

Particle method



Gaining attention with improved computing power, but rarely applied to real city models for breach flood analysis

The particle method has attracted attention with recent improvements in computer processing power.

There are existing studies on surface flow caused by rainfall, and on river flood analysis using virtual city models.

However, very few studies have used an actual city model to analyze flooding after a river breach.

PLATEAU



3D city models provided by the Ministry of Land, Infrastructure, Transport and Tourism, which include topographic data and building data

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In Japan, the Ministry of Land, Infrastructure, Transport and Tourism, or MLIT, provides 3D city data for cities across the country as a model called PLATEAU.

PLATEAU data is in a format called CityGML, an international standard for 3D city models.

It includes not just the shape of the ground and buildings, but also extra information, such as building height and construction year.

Purpose

River inundation analysis using particle method with PLATEAU...

- **Analysis of various elements that are damage risks**
- **Study of effectiveness as river inundation analysis**
- **Study of affinity between PLATEAU and particle method**

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In this study, we use PLATEAU and the particle method to analyze flooding caused by a river breach.

We look at several factors related to damage risk.

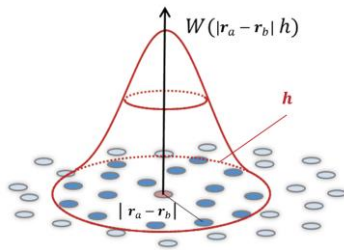
We also study how effective this method is for flood analysis, and how well PLATEAU and the particle method work together.

Smoothed Particle Hydrodynamics

Kernel function W

$$W(r, h) = \begin{cases} \alpha_D \left(1 - \frac{q}{2}\right)^4 (2q + 1) & 0 \leq q \leq 2 \\ 0 & q > 2 \end{cases}$$

$$\alpha_D = 21/16\pi h^3$$



$q[-]$: the non-dimensional distance between particles ($= r/h$)
 $r[\text{m}]$: the distance between any two given particles
 $h[\text{m}]$: the smoothing length

◀三富皓介, 高橋典之, 石田泰之: SPH法による流体解析を用いた市街地の水害評価に向けた検討-河川氾濫時にピロティ建物が市街地内の水流分布および周辺建物への水圧力に及ぼす影響-, 日本建築学会技術報告集, 第29巻, 第72号, pp. 671-676, 2023. より作成

Next, the approach.

This study uses a particle method called Smoothed Particle Hydrodynamics, or SPH.

In SPH, each particle has a kernel function, which gives more weight to nearby particles, and less weight to particles that are farther away.

Governing Equations

Momentum Conservation Equation, Continuity Equation, Equation of State

$$\frac{d\mathbf{u}_a}{dt} = - \sum_b m_a \left(\frac{p_b}{\rho_b^2} + \frac{p_a}{\rho_a^2} + \Pi_{ab} \right) \nabla W(|\mathbf{r}_a - \mathbf{r}_b|, h) + \mathbf{g}$$

$$\frac{d\rho_a}{dt} = \rho_a \sum_b \frac{m_b}{\rho_b} (\mathbf{u}_a - \mathbf{u}_b) \cdot \nabla W(|\mathbf{r}_a - \mathbf{r}_b|, h)$$

$$p = \frac{c_0 \rho_0}{\gamma} \left(\left(\frac{\rho}{\rho_0} \right)^\gamma - 1 \right) \quad \gamma = 7$$

$\mathbf{u}$ [m/s] : Velocity of a particle

η : Kinematic viscosity

$\mathbf{g}$ [m/s²] : Gravitational acceleration

W : Kernel function

Subscript : Indicates individual particles

ρ [kg/m³] : Density of a particle

p : Pressure of a particle

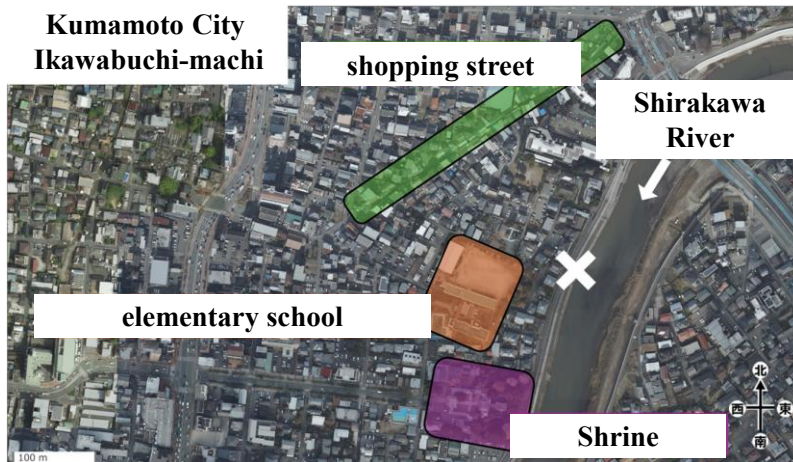
m [kg] : Mass of a particle

c_0 [m/s] : Speed of sound

ρ_0 [kg/m³] : Reference density

In the analysis, the fluid is treated as weakly compressible.
These governing equations describe momentum conservation, continuity, and the equation of state.

Target Area and Breach Locations



Created from 国土地理院電子国土基本図

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Next, the method.

The target area is Ikawabuchi-machi in Kumamoto City, on the right bank of the Shirakawa River.

Nearby landmarks include a shopping street, an elementary school, and a shrine.

Target Area and Breach Locations



FYI : 白川・緑川 水防災意識社会構築協議会 国土交通省 九州地方整備局

https://www.qsr.mlit.go.jp/kumamoto/site_files/file/river/shirakawa/mizubousaisikisyakaisaikoutikukyougikai/dai1kai/06siryou2.pdf

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This area was actually flooded during the heavy rain event in northern Kyushu, in July 2012.

Introduction

Approach

Method

Result

DualSPHysics (DSPH)



Free open source GPU-compatible software used in
various fluid dynamics analyses

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The software used for the analysis is DualSPHysics.
It is a free, open-source program that supports GPU computing, and is
used in many kinds of fluid analysis.

Analysis Parameters

Table : Analysis Settings Values

Analysis time	600s
Inter-particle distance r	0.5m
Smoothing length h	<0.5m
Inflow width (breach width)	45m
Inflow height (breach height)	2m
Inflow velocity	1m/s

For the settings, the analysis time is 600 seconds, and the distance between particles is 0.5 meters.

The breach is 45 meters wide and 2 meters high, with an inflow velocity of 1 meter per second.

Data used

Table : List of data

PLATEAU	Date title
Digital Elevation Model (5m)	493015_dem_6697_55_op_LOD1
Building Model (FY2022)	49301567_bldg_6697_op_LOD1 49301577_bldg_6697_op_LOD1



FYI : 国土交通省 PLATEAU Learning TOPIC1

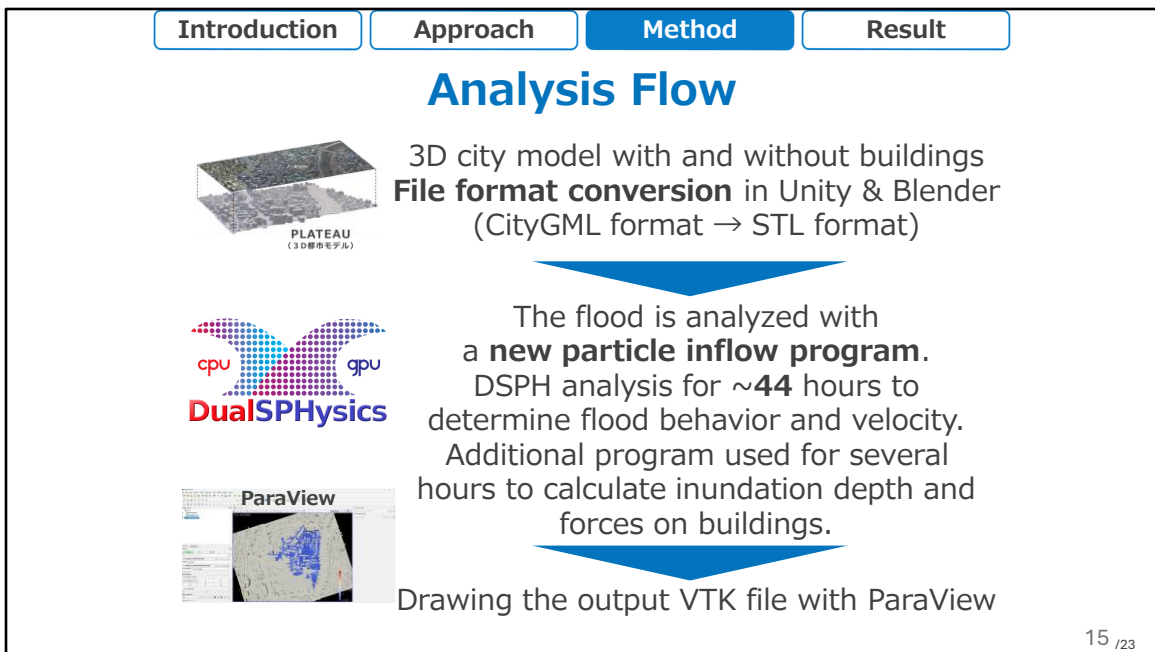
<https://www.mlit.go.jp/plateau/learning/tpc01-2/>

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The data used are PLATEAU's digital elevation model, and its building model.

For the buildings, we used LOD1 data, which represents each building as a simple box, with its footprint and height.

We ran the analysis twice: once without buildings, and once with buildings.



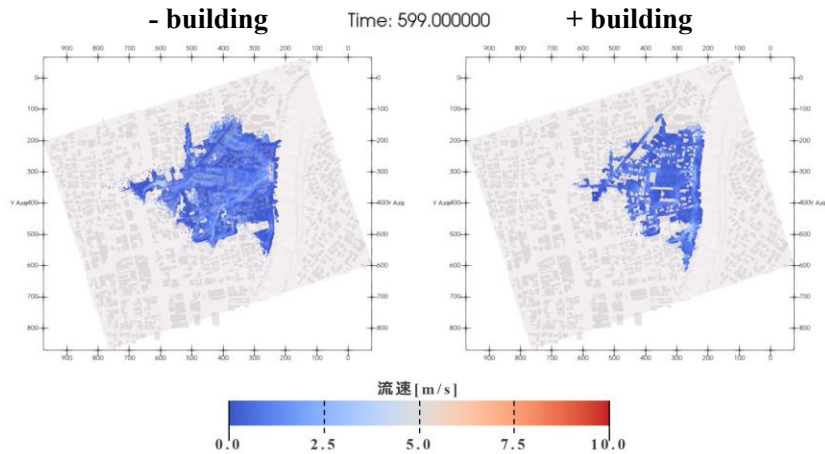
We converted the 3D city model, with and without buildings, from CityGML format to STL format, using Unity and Blender.

Then, using a new particle-inflow program, we ran the DSPH analysis for about 44 hours, to get the flood behavior and velocity of each particle.

After that, an additional program calculated the flood depth and the forces on buildings from these results, over several hours.

Finally, we visualized the results using ParaView.

Velocity



Now, the results.

Without buildings, the flood flows quickly, because there is nothing to block it.

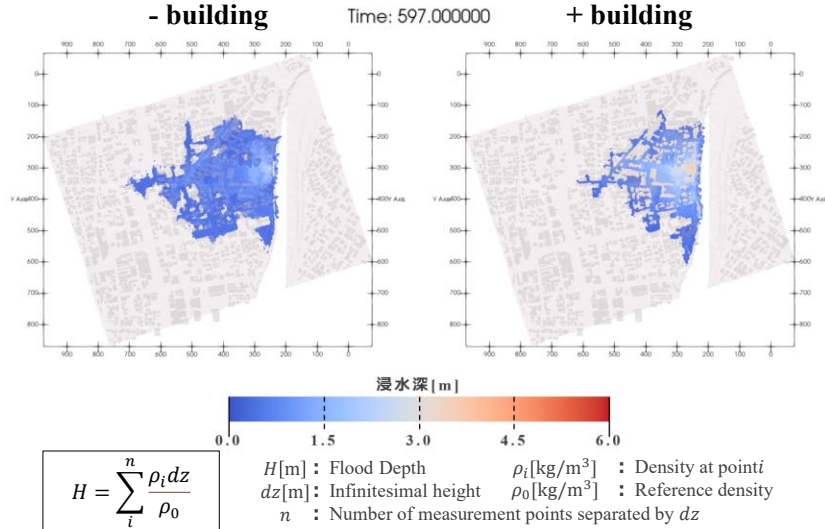
With buildings, the flow's momentum is weakened, resulting in a slower velocity.

Flood Flow



Looking more closely at the results with buildings, the flow swirls in some areas, and becomes faster in narrow gaps between buildings.

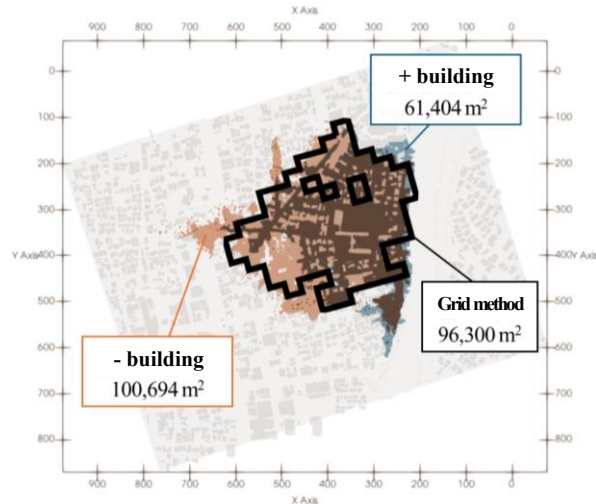
Flood Depth



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The flood depth is calculated from the density of the particles.
Comparing the two cases, the flood depth is higher with buildings than without.

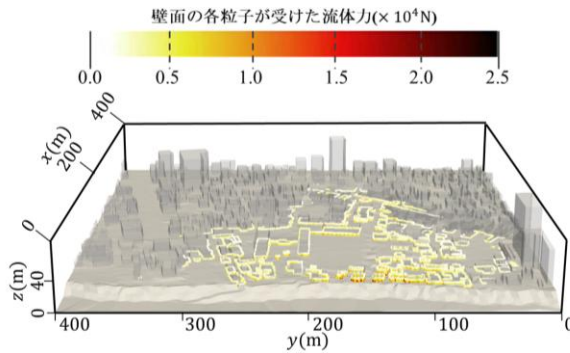
Comparison of Inundation Areas



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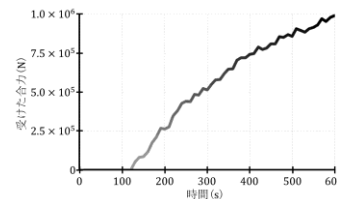
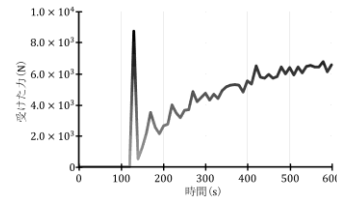
We compared the flooded area of each case with the grid method. Without buildings, the result was close to the grid method. With buildings, the flooded area was about 64% smaller, because the buildings blocked the horizontal flow, and changed its direction vertically.

Forces Acting on Buildings



$$F = m \sum \frac{du_a}{dt}$$

$F[\text{N}]$: Forces acting on buildings
 $m[\text{kg}]$: Mass



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The left figure shows where fluid forces are acting on buildings across the whole analysis area, at 600 seconds — the end of the simulation.

The top-right graph then shows how the force changes over time, at a point where the flow directly hits a building wall.

The bottom-right graph shows the total force over time, for the wall that contains that point, reaching about 1,000 kilonewtons by 600 seconds.

By showing specific numbers like this, we believe we can evaluate risks such as building collapse in more detail.

Conclusion

River inundation analysis using particle method with PLATEAU...

- **High validity compared to the lattice method, as close results were obtained without buildings**
- **High affinity between PLATEAU and particle methods because results were obtained using only numerical elevation data and building models**

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Finally, a summary.

With the particle method, we were able to determine the flood flow's behavior — its velocity, depth, and inundation area — as well as its impact on buildings.

Without buildings, the results were close to the grid method, showing that this approach is effective for flood analysis.

Since we only needed elevation data and a building model to get these results, PLATEAU and the particle method appear to work well together.

Future perspective

- **Damage risk assessment for each building**
 - **Changes in flood flow from water entering buildings**
 - **Changes in flood paths from floating debris (e.g. vehicles, trees)**
- etc.

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Going forward, we plan to look at damage risk assessment for each building, changes in flood flow caused by water entering buildings, and changes in flood paths caused by floating debris such as cars and trees.

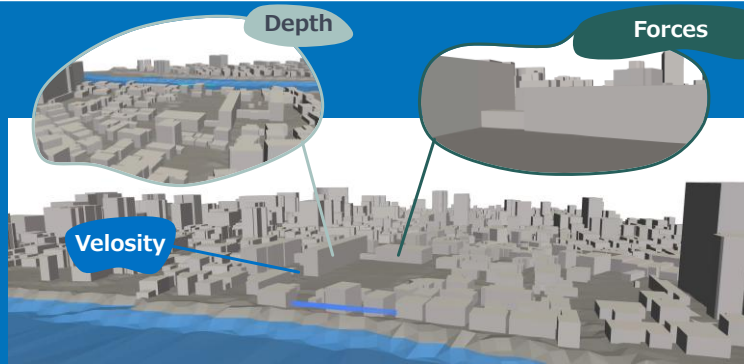
With further development, we believe a single analysis will be able to show a wider range of damage risks than in this study.

References

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- 2) 越塚誠一, 柴田和也, 室谷浩平：粒子法入門, pp. 19-32, 丸善出版, 2014.
- 3) Crespo AJC, et al., : Dual SPHysics | Open-source parallel CFD solver based on Smoothed Particle Hydrodynamics, Computer Physics Communications, 187, pp.204-216, 2015.
- 4) 国土交通省：PLATEAU, <https://www.mlit.go.jp/plateau/> (参照2024.12.12)
- 5) 国土地理院：電子国土基本図, <https://maps.gsi.go.jp> (参照2024.12.12)
- 6) 三富皓介, 高橋典之, 石田泰之：SPH 法による流体解析を用いた市街地の水害評価に向けた検討－河川氾濫時にピロティ建物が市街地内の水流分布および周辺建物への水圧力に及ぼす影響－, 日本建築学会技術報告集, 第29巻, 第72号, pp. 671-676, 2023.
- 7) Xin, Yan, LYE. Akihiko, N. and Sin Ying TAN.: Simulation of Overland Flow due to Rainfall using Smoothed Particle Hydrodynamics, Journal of Japan Society of Civil Engineers, Ser. B1 (Hydraulic Engineering), Vol. 78, No. 2, I_553-I_558, 2022.

These are the references used in this study.

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That concludes my presentation. Thank you for your attention.